# An energy budget

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Connect work, motion, heating and an honest model conclusion.

**SUCCESS CRITERIA - I can...**

1. Keep all terms in consistent energy units.
2. Distinguish mechanical-energy change from total-energy loss.
3. Report assumptions with the energy budget.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

An energy account begins with 15 J. It ends with 9 J motion energy and 3 J thermal energy. Find the unaccounted energy and suggest a next check.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $E_i + E_{in} = E_f + E_{out}$.
i/f = initial/final stored energy; in/out = transfers across the chosen system boundary. Account for every form.

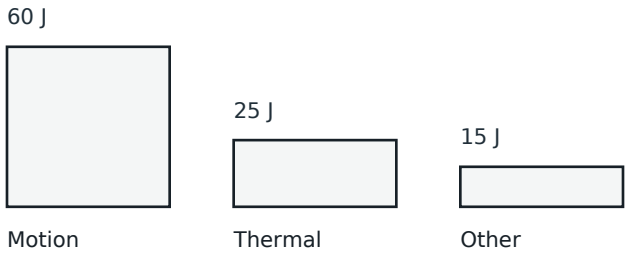
**Worked reasoning:** An input of 100 J gives 60 J useful motion, 25 J thermal transfer and 15 J sound/other transfers. The account balances because 60+25+15=100 J.

**Your turn:** Draw proportional energy arrows and calculate the useful fraction.

**Marzano check - circle:** ① ② ③

## 03 Look and explain

A modeled system starts with 100 J. It ends with 60 J motion energy, 25 J thermal energy and 15 J sound/deformation. Draw an energy account and check the total.




Bar height represents energy. Read the value and unit above each bar.

**Marzano check - circle:** ① ② ③

## 04 Build the solution

A 2 kg cart starts at 4 m/s and receives 8 J work. Find initial and final kinetic energy if no losses. Then include 5 J transferred to thermal energy.

**Marzano check - circle:** ① ② ③

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Continue down each column.

### 05 Use the idea

What information is missing if an impact report gives only total kinetic energy and says all of it heats the ground?

**Marzano check - circle:** 1 2 3

### 06 Choose a method

Gravitational potential energy changes from -24 J to -4 J. Find ΔU and state whether energy is gained or lost.

**Marzano check - circle:** 1 2 3

### 07 Exit check

Input energy 40 J yields 30 J useful mechanical energy. Find other energy and useful fraction.

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

Improve useful output to 75 J without changing input. Design two possible new thermal/other accounts and explain why data would be needed to choose one.

**Marzano check - circle:** 1 2 3